

Ablation Study Report

MambaOut-Tiny on ImageNet-100

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1. Experiment Setup

In this project, we study the performance of the MambaOut-Tiny model on the ImageNet-100 dataset.

The purpose is to understand how different hyperparameters and internal model components influence the final accuracy.

Model: MambaOut-Tiny

Dataset: ImageNet-100 (train and validation split prepared manually)

Hardware: HKUST SuperPOD, 1× NVIDIA H800 GPU

Batch size: 64

Image size: 224x224

Training length: 20 epochs for all settings

Baseline hyperparameters:

drop-path = 0.1

mixup = 0.8

cutmix = 1.0

weight decay = 0.05

learning rate = 3e-4

This baseline is used as the reference for all ablation experiments.

2. Training Logs

Below are the raw training logs for each experiment.

In the PDF, only the end of each log are shown.

2.1 Baseline (drop-path=0.1, mixup=0.8, cutmix=1.0)

Best Top-1 Accuracy: 72.60%

Log file: baseline_tiny_output.txt

```
Test: [ 200/202] Time: 0.017 (0.036) Loss: 0.8906 (1.1983) Acc@1: 82.8125 (72.5435) Acc@5: 93.7500 (92.1486)
Test: [ 202/202] Time: 0.004 (0.036) Loss: 0.5454 (1.1958) Acc@1: 90.0000 (72.6001) Acc@5: 100.0000 (92.1704)
Current checkpoints:
('outputs/tiny_baseline_dp0.1/20251117-234608-mambaout_tiny-224/checkpoint-19.pth.tar', 72.60009275003864)
('outputs/tiny_baseline_dp0.1/20251117-234608-mambaout_tiny-224/checkpoint-18.pth.tar', 72.2213634255681)
('outputs/tiny_baseline_dp0.1/20251117-234608-mambaout_tiny-224/checkpoint-17.pth.tar', 71.14700881125367)
('outputs/tiny_baseline_dp0.1/20251117-234608-mambaout_tiny-224/checkpoint-16.pth.tar', 68.87463286443035)
('outputs/tiny_baseline_dp0.1/20251117-234608-mambaout_tiny-224/checkpoint-15.pth.tar', 68.38769516153965)
('outputs/tiny_baseline_dp0.1/20251117-234608-mambaout_tiny-224/checkpoint-14.pth.tar', 66.64863193692997)
('outputs/tiny_baseline_dp0.1/20251117-234608-mambaout_tiny-224/checkpoint-13.pth.tar', 64.84000618333592)
('outputs/tiny_baseline_dp0.1/20251117-234608-mambaout_tiny-224/checkpoint-12.pth.tar', 62.0111300046375)
('outputs/tiny_baseline_dp0.1/20251117-234608-mambaout_tiny-224/checkpoint-11.pth.tar', 60.650796104498376)
('outputs/tiny_baseline_dp0.1/20251117-234608-mambaout_tiny-224/checkpoint-10.pth.tar', 56.86350285979286)

Train: 20 [ 0/1819 ( 0%)] Loss: 3.503 (3.50) Time: 0.326s, 196.44/s (0.326s, 196.44/s) LR: 8.250e-05 Data: 0.271 (0.271)
Train: 20 [ 50/1819 ( 3%)] Loss: 2.749 (2.97) Time: 0.059s, 1086.61/s (0.066s, 966.72/s) LR: 8.250e-05 Data: 0.004 (0.011)
Train: 20 [ 100/1819 ( 6%)] Loss: 2.135 (2.98) Time: 0.060s, 1074.98/s (0.063s, 1019.60/s) LR: 8.250e-05 Data: 0.005 (0.008)
Train: 20 [ 150/1819 ( 8%)] Loss: 3.021 (3.03) Time: 0.060s, 1074.73/s (0.062s, 1039.10/s) LR: 8.250e-05 Data: 0.005 (0.007)
Train: 20 [ 200/1819 (11%)] Loss: 2.395 (3.01) Time: 0.059s, 1091.83/s (0.061s, 1050.67/s) LR: 8.250e-05 Data: 0.004 (0.006)
Train: 20 [ 250/1819 (14%)] Loss: 3.844 (3.00) Time: 0.060s, 1073.34/s (0.061s, 1057.06/s) LR: 8.250e-05 Data: 0.005 (0.006)
^C*** Best metric: 72.60009275003864 (epoch 19)
```

2.2 Drop-Path = 0.0

Best Top-1 Accuracy: 63.63%

Log file: dp0.0_tiny_output.txt

```
Train: 19 [1800/1819 ( 99%)] Loss: 2.896 (3.38) Time: 0.057s, 1117.42/s (0.058s, 1112.74/s) LR: 1.000e-05 Data: 0.004 (0.004)
Train: 19 [1818/1819 (100%)] Loss: 2.621 (3.38) Time: 0.053s, 1209.65/s (0.058s, 1112.82/s) LR: 1.000e-05 Data: 0.000 (0.004)
Test: [ 0/202] Time: 0.288 (0.288) Loss: 0.7515 (0.7515) Acc@1: 92.1875 (92.1875) Acc@5: 95.3125 (95.3125)
Test: [ 50/202] Time: 0.017 (0.038) Loss: 1.5547 (1.2366) Acc@1: 68.7500 (72.5797) Acc@5: 84.3750 (92.3100)
Test: [ 100/202] Time: 0.017 (0.037) Loss: 1.8486 (1.5340) Acc@1: 56.2500 (64.5421) Acc@5: 82.8125 (87.2370)
Test: [ 150/202] Time: 0.018 (0.036) Loss: 1.3105 (1.6955) Acc@1: 59.3750 (59.5820) Acc@5: 95.3125 (84.9234)
Test: [ 200/202] Time: 0.017 (0.035) Loss: 1.4697 (1.5693) Acc@1: 70.3125 (63.6272) Acc@5: 87.5000 (86.6604)
Test: [ 202/202] Time: 0.004 (0.035) Loss: 1.6436 (1.5687) Acc@1: 60.0000 (63.6343) Acc@5: 90.0000 (86.6672)
Current checkpoints:
('outputs/tiny_dp0.0/20251118-011452-mambaout_tiny-224/checkpoint-19.pth.tar', 63.634255680939866)
('outputs/tiny_dp0.0/20251118-011452-mambaout_tiny-224/checkpoint-18.pth.tar', 63.25552635646932)
('outputs/tiny_dp0.0/20251118-011452-mambaout_tiny-224/checkpoint-17.pth.tar', 62.714484464368525)
('outputs/tiny_dp0.0/20251118-011452-mambaout_tiny-224/checkpoint-16.pth.tar', 62.498067707528214)
('outputs/tiny_dp0.0/20251118-011452-mambaout_tiny-224/checkpoint-15.pth.tar', 61.85654660689442)
('outputs/tiny_dp0.0/20251118-011452-mambaout_tiny-224/checkpoint-14.pth.tar', 61.841088267120114)
('outputs/tiny_dp0.0/20251118-011452-mambaout_tiny-224/checkpoint-13.pth.tar', 61.03725459885608)
('outputs/tiny_dp0.0/20251118-011452-mambaout_tiny-224/checkpoint-12.pth.tar', 60.782191992579996)
('outputs/tiny_dp0.0/20251118-011452-mambaout_tiny-224/checkpoint-11.pth.tar', 60.29525428968929)
('outputs/tiny_dp0.0/20251118-011452-mambaout_tiny-224/checkpoint-10.pth.tar', 59.23635801514917)
*** Best metric: 63.634255680939866 (epoch 19)
```

2.3 Drop-Path = 0.2

Best Top-1 Accuracy: 60.79%

Log file: dp0.2_tiny_output.txt

```
Train: 19 [1800/1819 ( 99%)] Loss: 3.058 (3.48) Time: 0.059s, 1083.48/s (0.060s, 1063.83/s) LR: 1.000e-05 Data: 0.004 (0.005)
Train: 19 [1818/1819 (100%)] Loss: 2.754 (3.48) Time: 0.055s, 1164.59/s (0.060s, 1063.95/s) LR: 1.000e-05 Data: 0.000 (0.005)
Test: [ 0/202] Time: 0.327 (0.327) Loss: 0.6958 (0.6958) Acc@1: 87.5000 (87.5000) Acc@5: 96.8750 (96.8750)
Test: [ 50/202] Time: 0.017 (0.048) Loss: 1.5703 (1.3219) Acc@1: 65.6250 (69.5466) Acc@5: 84.3750 (91.3909)
Test: [100/202] Time: 0.018 (0.043) Loss: 1.8447 (1.6317) Acc@1: 56.2500 (61.5099) Acc@5: 82.8125 (85.9684)
Test: [150/202] Time: 0.017 (0.044) Loss: 1.3535 (1.7993) Acc@1: 62.5000 (56.5604) Acc@5: 95.3125 (83.3195)
Test: [200/202] Time: 0.017 (0.042) Loss: 1.6055 (1.6677) Acc@1: 68.7500 (60.7898) Acc@5: 87.5000 (85.1368)
Test: [202/202] Time: 0.004 (0.041) Loss: 1.7676 (1.6670) Acc@1: 60.0000 (60.7899) Acc@5: 90.0000 (85.1523)
Current checkpoints:
('outputs/tiny_dp0.2/20251118-020411-mambaout_tiny-224/checkpoint-19.pth.tar', 60.78992116246715)
('outputs/tiny_dp0.2/20251118-020411-mambaout_tiny-224/checkpoint-18.pth.tar', 60.24887927036636)
('outputs/tiny_dp0.2/20251118-020411-mambaout_tiny-224/checkpoint-17.pth.tar', 59.785129077137114)
('outputs/tiny_dp0.2/20251118-020411-mambaout_tiny-224/checkpoint-16.pth.tar', 59.15906631627763)
('outputs/tiny_dp0.2/20251118-020411-mambaout_tiny-224/checkpoint-15.pth.tar', 59.10496212706755)
('outputs/tiny_dp0.2/20251118-020411-mambaout_tiny-224/checkpoint-14.pth.tar', 58.77260782191993)
('outputs/tiny_dp0.2/20251118-020411-mambaout_tiny-224/checkpoint-13.pth.tar', 58.12335755139898)
('outputs/tiny_dp0.2/20251118-020411-mambaout_tiny-224/checkpoint-12.pth.tar', 57.52821147008811)
('outputs/tiny_dp0.2/20251118-020411-mambaout_tiny-224/checkpoint-11.pth.tar', 57.16494048539187)
('outputs/tiny_dp0.2/20251118-020411-mambaout_tiny-224/checkpoint-10.pth.tar', 56.9407945586644)
*** Best metric: 60.78992116246715 (epoch 19)
```

2.4 mixup = 0, cutmix = 1.0

Best Top-1 Accuracy: 61.95%

Log file: mix0_cut1_tiny_output.txt

```
Train: 19 [1800/1819 ( 99%)] Loss: 3.246 (3.40) Time: 0.060s, 1071.45/s (0.060s, 1061.91/s) LR: 1.000e-05 Data: 0.005 (0.005)
Train: 19 [1818/1819 (100%)] Loss: 2.749 (3.40) Time: 0.055s, 1164.25/s (0.060s, 1061.97/s) LR: 1.000e-05 Data: 0.000 (0.005)
Test: [ 0/202] Time: 0.471 (0.471) Loss: 0.7324 (0.7324) Acc@1: 89.0625 (89.0625) Acc@5: 95.3125 (95.3125)
Test: [ 50/202] Time: 0.017 (0.044) Loss: 1.5645 (1.2532) Acc@1: 65.6250 (71.2623) Acc@5: 85.9375 (91.8199)
Test: [100/202] Time: 0.018 (0.040) Loss: 1.5547 (1.5658) Acc@1: 62.5000 (62.6083) Acc@5: 90.6250 (86.6027)
Test: [150/202] Time: 0.018 (0.041) Loss: 1.2744 (1.7288) Acc@1: 65.6250 (57.8642) Acc@5: 96.8750 (84.2094)
Test: [200/202] Time: 0.017 (0.039) Loss: 1.3926 (1.6031) Acc@1: 71.8750 (61.9325) Acc@5: 87.5000 (85.8520)
Test: [202/202] Time: 0.004 (0.039) Loss: 1.5898 (1.6020) Acc@1: 50.0000 (61.9493) Acc@5: 100.0000 (85.8943)
Current checkpoints:
('outputs/tiny_mix0_cut1/20251118-024948-mambaout_tiny-224/checkpoint-19.pth.tar', 61.94929664554027)
('outputs/tiny_mix0_cut1/20251118-024948-mambaout_tiny-224/checkpoint-18.pth.tar', 61.655588189828414)
('outputs/tiny_mix0_cut1/20251118-024948-mambaout_tiny-224/checkpoint-17.pth.tar', 61.0681712784047)
('outputs/tiny_mix0_cut1/20251118-024948-mambaout_tiny-224/checkpoint-16.pth.tar', 60.52712938630391)
('outputs/tiny_mix0_cut1/20251118-024948-mambaout_tiny-224/checkpoint-15.pth.tar', 60.24115010047921)
('outputs/tiny_mix0_cut1/20251118-024948-mambaout_tiny-224/checkpoint-14.pth.tar', 59.777390907248965)
('outputs/tiny_mix0_cut1/20251118-024948-mambaout_tiny-224/checkpoint-13.pth.tar', 59.35229556345649)
('outputs/tiny_mix0_cut1/20251118-024948-mambaout_tiny-224/checkpoint-12.pth.tar', 58.63348276395115)
('outputs/tiny_mix0_cut1/20251118-024948-mambaout_tiny-224/checkpoint-11.pth.tar', 58.540732725305304)
('outputs/tiny_mix0_cut1/20251118-024948-mambaout_tiny-224/checkpoint-10.pth.tar', 57.50502396042665)
*** Best metric: 61.94929664554027 (epoch 19)
```

2.5 mixup = 0.8, cutmix = 0

Best Top-1 Accuracy: 62.75%

Log file: mix0.8_cut0_tiny_output.txt

```
Train: 19 [1800/1819 ( 99%)] Loss: 2.400 (3.34) Time: 0.058s, 1109.40/s (0.059s, 1092.19/s) LR: 1.000e-05 Data: 0.004 (0.004)
Train: 19 [1818/1819 (100%)] Loss: 4.013 (3.33) Time: 0.054s, 1190.34/s (0.059s, 1092.29/s) LR: 1.000e-05 Data: 0.000 (0.004)
Test: [ 0/202] Time: 0.291 (0.291) Loss: 0.7432 (0.7432) Acc@1: 87.5000 (87.5000) Acc@5: 93.7500 (93.7500)
Test: [ 50/202] Time: 0.017 (0.030) Loss: 1.4131 (1.2131) Acc@1: 60.9375 (71.6605) Acc@5: 79.6875 (91.8199)
Test: [100/202] Time: 0.016 (0.035) Loss: 1.8027 (1.5162) Acc@1: 53.1250 (63.8304) Acc@5: 85.9375 (86.8348)
Test: [150/202] Time: 0.018 (0.036) Loss: 1.3574 (1.6820) Acc@1: 56.2500 (58.8266) Acc@5: 96.8750 (84.3647)
Test: [200/202] Time: 0.016 (0.035) Loss: 1.3643 (1.5551) Acc@1: 65.6250 (62.7332) Acc@5: 85.9375 (86.0308)
Test: [202/202] Time: 0.004 (0.035) Loss: 1.2393 (1.5531) Acc@1: 60.0000 (62.7531) Acc@5: 100.0000 (86.0798)
Current checkpoints:
('outputs/tiny_mix0.8_cut0/20251118-201416-mambaout_tiny-224/checkpoint-19.pth.tar', 62.7531303138043)
('outputs/tiny_mix0.8_cut0/20251118-201416-mambaout_tiny-224/checkpoint-18.pth.tar', 62.08069253362189)
('outputs/tiny_mix0.8_cut0/20251118-201416-mambaout_tiny-224/checkpoint-17.pth.tar', 62.003400834750344)
('outputs/tiny_mix0.8_cut0/20251118-201416-mambaout_tiny-224/checkpoint-16.pth.tar', 61.47781728242387)
('outputs/tiny_mix0.8_cut0/20251118-201416-mambaout_tiny-224/checkpoint-15.pth.tar', 61.21502550626063)
('outputs/tiny_mix0.8_cut0/20251118-201416-mambaout_tiny-224/checkpoint-14.pth.tar', 60.92131705054877)
('outputs/tiny_mix0.8_cut0/20251118-201416-mambaout_tiny-224/checkpoint-13.pth.tar', 60.08656670273613)
('outputs/tiny_mix0.8_cut0/20251118-201416-mambaout_tiny-224/checkpoint-12.pth.tar', 59.76194156747565)
('outputs/tiny_mix0.8_cut0/20251118-201416-mambaout_tiny-224/checkpoint-11.pth.tar', 59.63827484928119)
('outputs/tiny_mix0.8_cut0/20251118-201416-mambaout_tiny-224/checkpoint-10.pth.tar', 58.76487865203277)
*** Best metric: 62.7531303138043 (epoch 19)
```

2.6 mixup = 0, cutmix = 0

Best Top-1 Accuracy: 76.80%

Log file: mix0_cut0_tiny_output.txt

```
Train: 19 [1800/1819 ( 99%)] Loss: 1.840 (1.90) Time: 0.060s, 1068.77/s (0.060s, 1062.08/s) LR: 9.602e-05 Data: 0.005 (0.005)
Train: 19 [1818/1819 (100%)] Loss: 1.770 (1.90) Time: 0.055s, 1167.01/s (0.060s, 1062.12/s) LR: 9.602e-05 Data: 0.000 (0.005)
Test: [ 0/202] Time: 0.317 (0.317) Loss: 0.3325 (0.3325) Acc@1: 93.7500 (93.7500) Acc@5: 98.4375 (98.4375)
Test: [ 50/202] Time: 0.018 (0.043) Loss: 0.6938 (0.6195) Acc@1: 84.3750 (84.9877) Acc@5: 96.8750 (96.5380)
Test: [100/202] Time: 0.017 (0.040) Loss: 0.9824 (0.8253) Acc@1: 73.4375 (79.0068) Acc@5: 90.6250 (94.7246)
Test: [150/202] Time: 0.017 (0.041) Loss: 0.8560 (0.9767) Acc@1: 59.3750 (74.0273) Acc@5: 98.4375 (93.4396)
Test: [200/202] Time: 0.018 (0.039) Loss: 0.6699 (0.8858) Acc@1: 85.9375 (76.7646) Acc@5: 98.4375 (94.3097)
Test: [202/202] Time: 0.004 (0.039) Loss: 0.2737 (0.8840) Acc@1: 90.0000 (76.8048) Acc@5: 100.0000 (94.3191)
Current checkpoints:
('outputs/tiny_mix0_cut0/20251118-034128-mambaout_tiny-224/checkpoint-19.pth.tar', 76.80476116865049)
('outputs/tiny_mix0_cut0/20251118-034128-mambaout_tiny-224/checkpoint-18.pth.tar', 75.8076982532076)
('outputs/tiny_mix0_cut0/20251118-034128-mambaout_tiny-224/checkpoint-17.pth.tar', 75.08115628381512)
('outputs/tiny_mix0_cut0/20251118-034128-mambaout_tiny-224/checkpoint-16.pth.tar', 73.84448910187047)
('outputs/tiny_mix0_cut0/20251118-034128-mambaout_tiny-224/checkpoint-15.pth.tar', 72.75467614778172)
('outputs/tiny_mix0_cut0/20251118-034128-mambaout_tiny-224/checkpoint-14.pth.tar', 71.7498840624517)
('outputs/tiny_mix0_cut0/20251118-034128-mambaout_tiny-224/checkpoint-13.pth.tar', 70.42046684186118)
('outputs/tiny_mix0_cut0/20251118-034128-mambaout_tiny-224/checkpoint-12.pth.tar', 68.82825784510743)
('outputs/tiny_mix0_cut0/20251118-034128-mambaout_tiny-224/checkpoint-11.pth.tar', 67.90075745864894)
('outputs/tiny_mix0_cut0/20251118-034128-mambaout_tiny-224/checkpoint-10.pth.tar', 65.07961044983769)
*** Best metric: 76.80476116865049 (epoch 19)
```

2.7 disable-gated-multiplication

Best Top-1 Accuracy: 43.71%

Log file: disable_gate_output.txt

```
Train: 19 [1800/1819 ( 99%)] Loss: 3.496 (3.81) Time: 0.057s, 1122.15/s (0.058s, 1102.89/s) LR: 1.000e-05 Data: 0.004 (0.005)
Train: 19 [1818/1819 (100%)] Loss: 3.339 (3.81) Time: 0.053s, 1210.12/s (0.058s, 1102.90/s) LR: 1.000e-05 Data: 0.000 (0.005)
Test: [ 0/202] Time: 0.341 (0.341) Loss: 0.9795 (0.9795) Acc@1: 85.9375 (85.9375) Acc@5: 92.1875 (92.1875)
Test: [ 50/202] Time: 0.018 (0.046) Loss: 2.0566 (1.9809) Acc@1: 56.2500 (53.3088) Acc@5: 76.5625 (83.3027)
Test: [100/202] Time: 0.018 (0.044) Loss: 2.5000 (2.3309) Acc@1: 46.8750 (44.6318) Acc@5: 75.0000 (74.8453)
Test: [150/202] Time: 0.017 (0.044) Loss: 1.8135 (2.4731) Acc@1: 50.0000 (40.0662) Acc@5: 90.6250 (71.7819)
Test: [200/202] Time: 0.017 (0.042) Loss: 2.5410 (2.3534) Acc@1: 34.3750 (43.7889) Acc@5: 75.0000 (73.9972)
Test: [202/202] Time: 0.004 (0.042) Loss: 2.5547 (2.3545) Acc@1: 20.0000 (43.7162) Acc@5: 80.0000 (73.9991)
Current checkpoints:
('outputs/tiny_disable_gate/20251118-170234-mambaout_tiny-224/checkpoint-19.pth.tar', 43.7161848817437)
('outputs/tiny_disable_gate/20251118-170234-mambaout_tiny-224/checkpoint-18.pth.tar', 43.437934765806155)
('outputs/tiny_disable_gate/20251118-170234-mambaout_tiny-224/checkpoint-17.pth.tar', 42.81960117483382)
('outputs/tiny_disable_gate/20251118-170234-mambaout_tiny-224/checkpoint-16.pth.tar', 42.29401762250734)
('outputs/tiny_disable_gate/20251118-170234-mambaout_tiny-224/checkpoint-15.pth.tar', 42.17808007420003)
('outputs/tiny_disable_gate/20251118-170234-mambaout_tiny-224/checkpoint-14.pth.tar', 41.35105889627454)
('outputs/tiny_disable_gate/20251118-170234-mambaout_tiny-224/checkpoint-13.pth.tar', 40.91822538259391)
('outputs/tiny_disable_gate/20251118-170234-mambaout_tiny-224/checkpoint-12.pth.tar', 40.454475189364665)
('outputs/tiny_disable_gate/20251118-170234-mambaout_tiny-224/checkpoint-11.pth.tar', 40.39264183026743)
('outputs/tiny_disable_gate/20251118-170234-mambaout_tiny-224/checkpoint-10.pth.tar', 39.33374555572731)
*** Best metric: 43.7161848817437 (epoch 19)
```

2.8 disable-conv-branch

Best Top-1 Accuracy: 53.67%

Log file: disable_conv_output.txt

```
Train: 19 [1800/1819 ( 99%)] Loss: 3.742 (3.61) Time: 0.050s, 1286.11/s (0.051s, 1253.42/s) LR: 1.000e-05 Data: 0.004 (0.005)
Train: 19 [1818/1819 (100%)] Loss: 3.934 (3.60) Time: 0.045s, 1415.95/s (0.051s, 1253.79/s) LR: 1.000e-05 Data: 0.000 (0.005)
Test: [ 0/202] Time: 0.286 (0.286) Loss: 0.9121 (0.9121) Acc@1: 87.5000 (87.5000) Acc@5: 95.3125 (95.3125)
Test: [ 50/202] Time: 0.015 (0.038) Loss: 1.6660 (1.6134) Acc@1: 65.6250 (62.5613) Acc@5: 87.5000 (88.0821)
Test: [100/202] Time: 0.016 (0.037) Loss: 2.1211 (1.9264) Acc@1: 48.4375 (54.7494) Acc@5: 81.2500 (82.0080)
Test: [150/202] Time: 0.016 (0.038) Loss: 1.7031 (2.0829) Acc@1: 53.1250 (50.1656) Acc@5: 89.0625 (78.6217)
Test: [200/202] Time: 0.015 (0.036) Loss: 1.9453 (1.9754) Acc@1: 54.6875 (53.6847) Acc@5: 82.8125 (80.3094)
Test: [202/202] Time: 0.003 (0.036) Loss: 2.0254 (1.9750) Acc@1: 40.0000 (53.6714) Acc@5: 90.0000 (80.3525)
Current checkpoints:
('outputs/tiny_disable_conv/20251118-174930-mambaout_tiny-224/checkpoint-19.pth.tar', 53.67135569639821)
('outputs/tiny_disable_conv/20251118-174930-mambaout_tiny-224/checkpoint-18.pth.tar', 52.49652187355078)
('outputs/tiny_disable_conv/20251118-174930-mambaout_tiny-224/checkpoint-17.pth.tar', 52.00185500077292)
('outputs/tiny_disable_conv/20251118-174930-mambaout_tiny-224/checkpoint-16.pth.tar', 51.63085484618952)
('outputs/tiny_disable_conv/20251118-174930-mambaout_tiny-224/checkpoint-15.pth.tar', 50.86566702736126)
('outputs/tiny_disable_conv/20251118-174930-mambaout_tiny-224/checkpoint-14.pth.tar', 50.64152110063379)
('outputs/tiny_disable_conv/20251118-174930-mambaout_tiny-224/checkpoint-13.pth.tar', 49.6289998454166)
('outputs/tiny_disable_conv/20251118-174930-mambaout_tiny-224/checkpoint-12.pth.tar', 48.63966609980875)
('outputs/tiny_disable_conv/20251118-174930-mambaout_tiny-224/checkpoint-11.pth.tar', 48.45416022569176)
('outputs/tiny_disable_conv/20251118-174930-mambaout_tiny-224/checkpoint-10.pth.tar', 47.3720822383676)
*** Best metric: 53.67135569639821 (epoch 19)
```

2.9 disable-residual-connection

Best Top-1 Accuracy: 1.8%

The model collapsed and did not learn stable features.

Log file: disable_residual_output.txt

```
Train: 19 [1200/1819 ( 66%)] Loss: 4.608 (4.61) Time: 0.056s, 1147.54/s (0.057s, 1119.95/s) LR: 1.000e-05 Data: 0.003 (0.004)
Train: 19 [1250/1819 ( 69%)] Loss: 4.604 (4.61) Time: 0.057s, 1121.78/s (0.057s, 1120.07/s) LR: 1.000e-05 Data: 0.004 (0.004)
Train: 19 [1300/1819 ( 72%)] Loss: 4.616 (4.61) Time: 0.057s, 1121.46/s (0.057s, 1120.22/s) LR: 1.000e-05 Data: 0.004 (0.004)
Train: 19 [1350/1819 ( 74%)] Loss: 4.603 (4.61) Time: 0.057s, 1129.09/s (0.057s, 1120.27/s) LR: 1.000e-05 Data: 0.004 (0.004)
Train: 19 [1400/1819 ( 77%)] Loss: 4.601 (4.61) Time: 0.057s, 1132.64/s (0.057s, 1120.36/s) LR: 1.000e-05 Data: 0.004 (0.004)
Train: 19 [1450/1819 ( 80%)] Loss: 4.604 (4.61) Time: 0.056s, 1133.05/s (0.057s, 1120.40/s) LR: 1.000e-05 Data: 0.003 (0.004)
Train: 19 [1500/1819 ( 83%)] Loss: 4.600 (4.61) Time: 0.058s, 1101.30/s (0.057s, 1120.31/s) LR: 1.000e-05 Data: 0.005 (0.004)
Train: 19 [1550/1819 ( 85%)] Loss: 4.608 (4.61) Time: 0.057s, 1121.63/s (0.057s, 1120.39/s) LR: 1.000e-05 Data: 0.004 (0.004)
Train: 19 [1600/1819 ( 88%)] Loss: 4.604 (4.61) Time: 0.057s, 1122.60/s (0.057s, 1120.41/s) LR: 1.000e-05 Data: 0.004 (0.004)
Train: 19 [1650/1819 ( 91%)] Loss: 4.602 (4.61) Time: 0.060s, 1073.56/s (0.057s, 1120.14/s) LR: 1.000e-05 Data: 0.005 (0.004)
Train: 19 [1700/1819 ( 94%)] Loss: 4.608 (4.61) Time: 0.057s, 1129.79/s (0.057s, 1120.00/s) LR: 1.000e-05 Data: 0.004 (0.004)
Train: 19 [1750/1819 ( 96%)] Loss: 4.607 (4.61) Time: 0.057s, 1126.17/s (0.057s, 1120.04/s) LR: 1.000e-05 Data: 0.004 (0.004)
Train: 19 [1800/1819 ( 99%)] Loss: 4.598 (4.61) Time: 0.057s, 1123.39/s (0.057s, 1120.04/s) LR: 1.000e-05 Data: 0.004 (0.004)
Train: 19 [1818/1819 (100%)] Loss: 4.600 (4.61) Time: 0.052s, 1224.02/s (0.057s, 1120.15/s) LR: 1.000e-05 Data: 0.000 (0.004)
Test: [ 0/202] Time: 0.291 (0.291) Loss: 4.5977 (4.5977) Acc@1: 0.0000 ( 0.0000) Acc@5: 0.0000 ( 0.0000)
Test: [ 50/202] Time: 0.016 (0.040) Loss: 4.5977 (4.6018) Acc@1: 0.0000 ( 0.0000) Acc@5: 0.0000 ( 7.9657)
Test: [100/202] Time: 0.016 (0.036) Loss: 4.6133 (4.6026) Acc@1: 0.0000 ( 2.0111) Acc@5: 0.0000 ( 8.0446)
Test: [150/202] Time: 0.016 (0.036) Loss: 4.6133 (4.6040) Acc@1: 0.0000 ( 1.3452) Acc@5: 0.0000 ( 6.7260)
Test: [200/202] Time: 0.031 (0.035) Loss: 4.6602 (4.6048) Acc@1: 0.0000 ( 1.0106) Acc@5: 0.0000 ( 5.0529)
Test: [202/202] Time: 0.004 (0.035) Loss: 4.6523 (4.6051) Acc@1: 0.0000 ( 1.0048) Acc@5: 0.0000 ( 5.0240)
*** Best metric: 1.808625753594064 (epoch 4)
```

3. Accuracy Curves

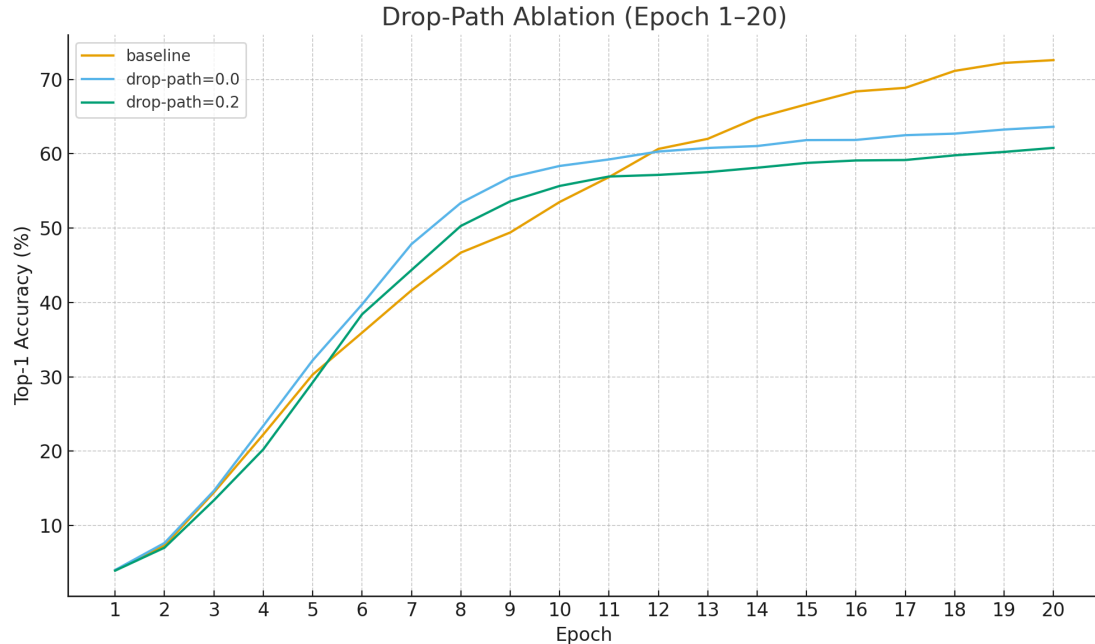
In this section, we include the accuracy curves for different ablations.

The x-axis is the epoch number from 1 to 20.

The y-axis is the Top-1 accuracy in percent.

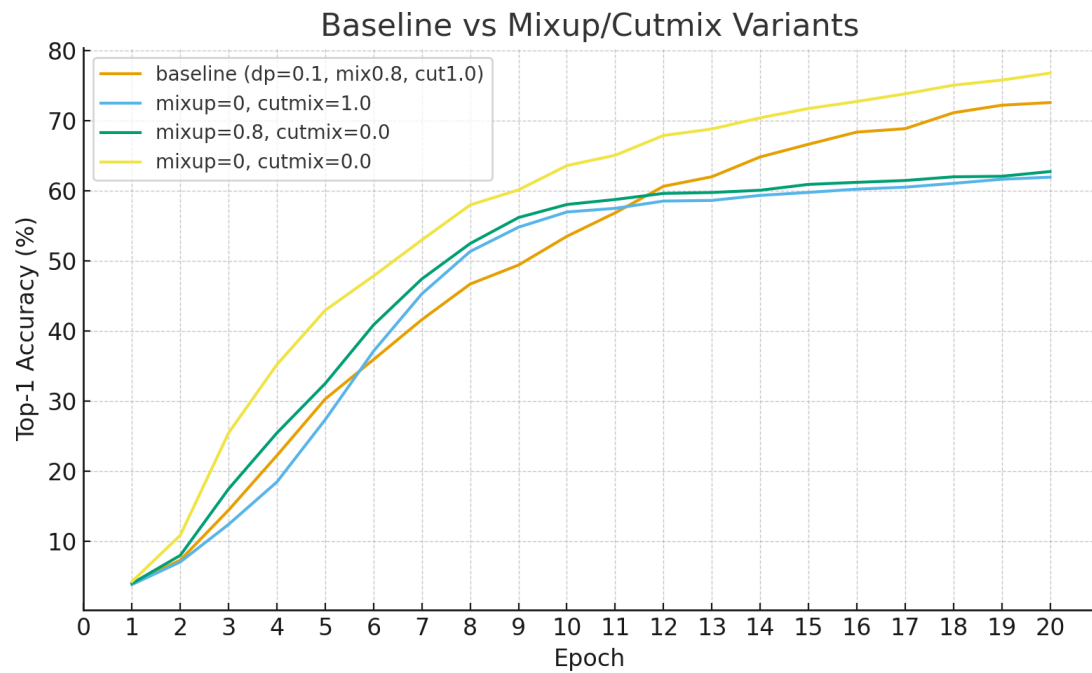
3.1 Drop-Path Comparison

Baseline (0.1) vs drop-path=0.0 vs drop-path=0.2



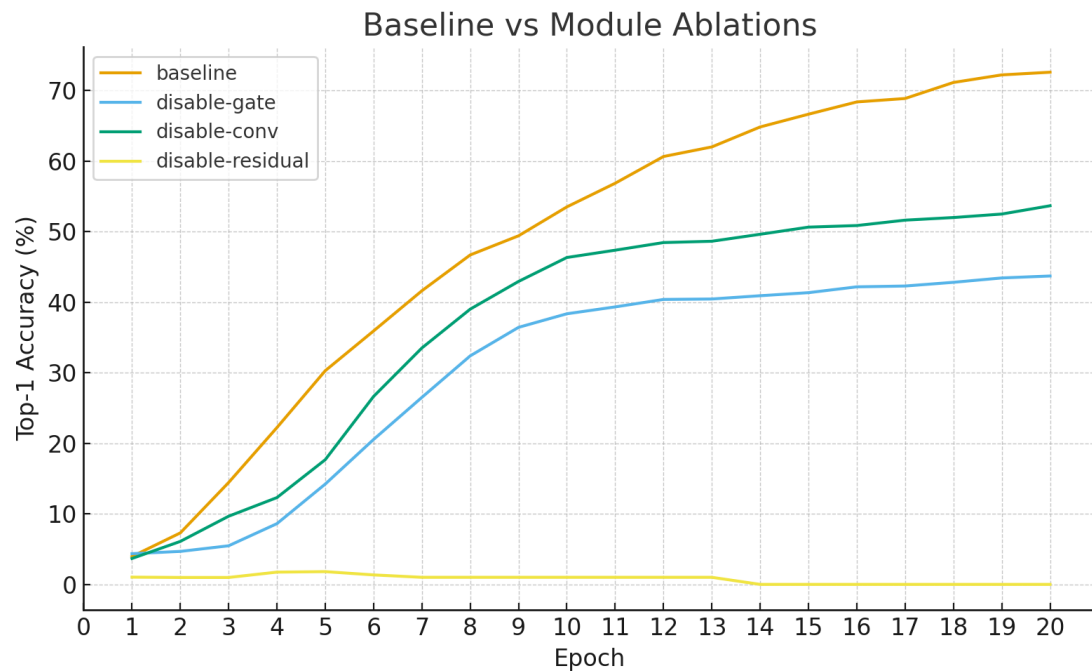
3.2 Mixup and Cutmix Comparison

Baseline vs (mix0 cut1v) vs (mix0.8 cut0) vs (mix0 cut0)



3.3 Module Ablations

Baseline vs disable-gate vs disable-conv vs disable-residual



4. Discussion and Findings

4.1 Drop-Path

The baseline value of 0.1 gives the best accuracy.

Setting drop-path to 0.0 removes regularization and causes lower accuracy.

Setting drop-path to 0.2 makes the regularization too strong, and the model becomes unstable in early epochs.

A moderate value is important for the MambaOut-Tiny model.

4.2 Mixup and Cutmix

Turning off both mixup and cutmix gives the highest accuracy among all experiments.

ImageNet-100 is much smaller than the full ImageNet-1k dataset, so strong mixed-label augmentation makes the training harder.

Without mixup and cutmix, the model learns cleaner decision boundaries and converges faster.

4.3 Module Ablations

disable-gated-multiplication

Accuracy drops by a large margin. This shows that the gate plays an important role in combining the two feature branches.

disable-conv-branch

Accuracy decreases by around 19 percent. The convolution branch still provides useful local information.

disable-residual-connection

The model fails almost completely, and accuracy stays around 1 percent.

Residual connections are essential for stable training in deep models.

5. Conclusion

We trained the MambaOut-Tiny model under different hyperparameters and different internal module settings.

The results show that moderate regularization and clean labels work better for ImageNet-100.

Some internal components, such as the residual connection, are necessary for stable training.

This ablation study helps us understand how the structure and training settings affect the model performance.